Capstone 14

AI Accessibility Study Aids

Making one NCERT chapter easier to read, hear and see

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Course	AI Skills Course, Topics 1-10 (Domain A: Education and Learning)
Source material	NCERT Class 10 Science, Chapter 11 "Electricity", Section 11.1 Electric Current and Circuit (pages 171-172 of the chapter PDF)
Report type	Progress report: work completed up to 20 October 2026
AI tools used	ChatGPT (free), Gemini (free), NotebookLM, Claude (free), ElevenLabs (free tier), Canva / image tool for the mind-map
Labelling	All AI-generated content in this report is labelled. Facts were checked against the NCERT text.

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1. Problem and objective

Project question: How can AI make study material easier to access for students who need simpler text, audio or visual formats?

Many students find dense textbook language hard to read, including students with dyslexia and students who are still learning the language of instruction. Others learn better by listening or by looking at pictures. One chapter section was turned into three formats and checked for lost or changed meaning.

Objective: Take one open resource section (NCERT Class 10 Science, Section 11.1) and produce (1) a simplified-language text, (2) a short audio version and (3) a visual summary. Compare three AI tools, check every output against the original, and record all errors.

Why this section: NCERT books are freely available. Section 11.1 has formulas, units, powers of ten and a worked example, so it is a good test of whether AI keeps meaning and numbers correct when simplifying.

2. Workflow and tools

Step	What was done	Tools
1. Choose source	Selected NCERT Class 10 Science, Ch. 11, Section 11.1	NCERT PDF
2. Simplify text	Same prompt run in three tools; outputs saved and labelled	ChatGPT, Gemini, NotebookLM
3. Verify	Each output compared line by line with the NCERT text; errors logged	NCERT text, manual check
4. Merge	Best parts combined, errors fixed by hand, final text written	Claude (assisted) + manual review
5. Audio script	Prompt 2 run in three tools on the final text; best parts merged	ChatGPT, Gemini, NotebookLM
6. Audio file	Final script converted to speech and listened to in full	ElevenLabs (free tier)
7. Mind-map	Prompt 3 run in three tools; outlines merged; picture designed	ChatGPT, Gemini, NotebookLM, image/design tool
8. Peer review	Google Form for 3 peers; results linked to Google Sheets	Google Forms, Google Sheets

Skills from Topics 1-9 used: AI foundations, prompt engineering, content creation, research and verification, image generation, audio and presentation tools, automation (Forms to Sheets), productivity workflow and responsible AI. That is more than the minimum of six.

3. Prompt library

Prompts were kept the same across tools so that the comparison is fair. Each prompt gave the tool a role, rules and an output format.

Prompt 1: simplified text

Role: accessibility specialist for students who find dense text hard to read. Rules: sentences under 15 words; explain each technical term in plain words the first time; keep all formulas and units exactly as in the original; do not add facts that are not in the text; short paragraphs; bulleted key-terms list at the end.

Prompt 2: audio narration script

Role: accessibility specialist writing audio study material. Rules: 2-3 minute script (about 300-400 words); calm, friendly tone; sentences under 15 words; spell out every formula and unit in words; say powers of ten in words (for example "ten to the power minus three"); no tables or special characters; do not add facts; keep all numbers exactly as given; end with a 3-point recap.

Prompt 3: mind-map outline

Role: accessibility specialist making visual study aids. Rules: one central idea; 5-6 branches; 2-3 short points per branch; each point under 8 words; keep formulas, units and numbers exactly; powers of ten with superscripts; do not add facts; plain text only.

Prompt improvement noted: NotebookLM did not follow Prompt 2 on the first try and replied with an offer to make an Audio Overview. It worked on the second try after the source was uploaded and only the rules were pasted. This is recorded as a tool-reliability finding.

4. Simplified text: original vs final

Original (NCERT Section 11.1), summary of its style: long sentences, terms such as "conductor", "cross-section", "conventionally" and "ampere" used without plain-word explanations, and the direction-of-current idea given in one dense paragraph.

Final simplified version (AI-assisted, verified against NCERT)

What is electric current?

Water flows in a river. This is a water current. In the same way, electric charge can flow through a conductor. A metallic wire is an example. This flow of charge is called an electric current.

What is an electric circuit?

A torch has cells and a bulb. The cells make charge flow through the bulb. The bulb then glows. A switch joins the cell and the bulb. The torch glows only when the switch is on. A continuous, closed path for electric current is an **electric circuit**. If the circuit breaks anywhere, the current stops. Then the bulb does not glow.

How do we express current?

Current is the amount of charge flowing through an area in unit time. In other words, it is the rate of flow of electric charge. Suppose a net charge Q flows across a cross-section of a conductor in time t. Then the current I is: **I = Q/t**

Which way does current flow?

In metal wires, electrons carry the charge. Electrons have negative charge. Electrons were not known when electricity was first studied. So people thought positive charges were flowing. They took that direction as the direction of current. By convention, current is opposite to the flow of electrons. Current flows from the positive terminal of the cell to the negative terminal.

Units

- The SI unit of charge is the **coulomb (C)**.
- One coulomb is the charge of nearly 6×10^{18} electrons.
- One electron has a negative charge of 1.6×10^{-19} C.
- The SI unit of current is the **ampere (A)**, named after Andre-Marie Ampere (1775-1836).
- One ampere means one coulomb of charge flows per second: **1 A = 1 C / 1 s**.
- 1 mA (milliampere) = 10^{-3} A and 1 μA (microampere) = 10^{-6} A.

Measuring current

An **ammeter** measures current in a circuit. It is always connected in series.

Worked example (NCERT Example 11.1)

A current of 0.5 A flows through a bulb filament for 10 minutes. Find the charge. Given: I = 0.5 A, t = 10 min = 600 s. Formula: Q = It. Q = 0.5 A × 600 s = **300 C**.

Key terms: Electric current (I): rate of flow of electric charge, in amperes. Electric circuit: a continuous, closed path for current. Electrons: particles in metal wires that carry negative charge. Coulomb (C): SI unit of charge. Ampere (A): SI unit of current. Ammeter: instrument that measures current, connected in series.

Honest limit: a few sentences are still slightly longer than the 15-word rule (for example the positive-to-negative terminal sentence). They were kept to avoid losing meaning.

5. Verification: error logs and tool comparison

Every output was compared line by line with the NCERT text. Severity: High = changes meaning or breaks a formula; Medium = lost precision or added content; Low = wording or tone.

5.1 Simplified text, Section 11.1

Tool	Main errors found	Severity
ChatGPT	Defined charge as "tiny charged particles moving through a wire" (not in NCERT). Exponents flattened (written as 10-3 with no superscript, so it reads like 10 minus 3). Added "energy means ability to do work". Units dropped in the worked example. Prompt text and file tags leaked into the output. LaTeX shows as raw code in documents.	High (charge, exponents)
NotebookLM	Called an electron "a charge" in key terms. Added "tiny". Dropped the worked example. Citation numbers and LaTeX fragments left in. Several sentences over 15 words. All formulas and exponents were correct.	Medium
Gemini	Added a wrong source tag ("ALLEN's") although the source was NCERT. Exponents and formulas garbled (6×1018). Many omissions: 1 A = 1 C/1 s, mA and µA, ammeter in series, worked example.	High (source tag, exponents)

A separate run of Gemini on Section 11.4 (Ohm's law) wrote "current increases when potential difference increases" instead of "directly proportional", and its formulas were corrupted into text like V=IRV = IRV=IR. It is recorded as an extra example of lost precision.

5.2 Comparison on the same section

Criterion	ChatGPT	NotebookLM	Gemini
Formula accuracy	Exponents broken, units dropped	All correct	Exponents and formulas corrupted
Added or invented facts	Several	Few	Few
Most serious error	Wrong definition of charge	Electron called a "charge"	Wrong source tag, garbled exponent
Completeness	Has example, ammeter, mA/µA	Most complete, no example	Thinnest
Readability	Good	Dense, long sentences	Good, short sentences

Finding: NotebookLM was the most accurate because it works from the uploaded source. ChatGPT and Gemini were easier to read but added content or broke formulas. No tool followed every rule. Human review was needed in every case.

5.3 Fixes applied in the final text

Error in tool output	Fix
ChatGPT: charge defined as "tiny particles moving"	Removed; only NCERT wording used

Error in tool output	Fix
ChatGPT: "energy means ability to do work"	Removed (not in NCERT)
ChatGPT and Gemini: flattened exponents	Rewritten as 10^{-3}, 10^{-6}, 6×10^{18}
ChatGPT: units dropped in example	Kept A and s in the calculation
NotebookLM: electron called a "charge"	Now "particles that carry negative charge"
NotebookLM and Gemini: example missing	Added Example 11.1
Gemini: "ALLEN's" source tag	Removed; source is NCERT only
Gemini: missing 1 A = 1 C/1 s, mA, µA, ammeter	Added back
All tools: citation numbers, LaTeX, leaked prompt text	Cleaned to plain text

Note: an early practice text written from memory (before the real NCERT text was available) also had gaps, such as describing voltage only as a "push". This showed why verification against the source is needed.